Quantitative Ecology Meets the High Street: Using Biodiversity Metrics to Assess Multiscale Retail Structure in Brussels

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De l'art de détourner des outils de l'écologie quantitative pour faire des trucs (intéressants) en géographie : application au commerce bruxellois

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Intuition

- La ville est une mosaïque d'activités (économiques)

- Comment appréhender cette complexité ?
  - Ville de proximité/Ville à X minutes
  - Mixité des fonctions

Quantifier la « diversité urbaine » (I)

- Traditionnellement, on utilise :
  - Une somme d'activités (calcul simple)
  - Une mesure d'entropie (calcul plus complexe)

- Entropie de Shannon (1948)

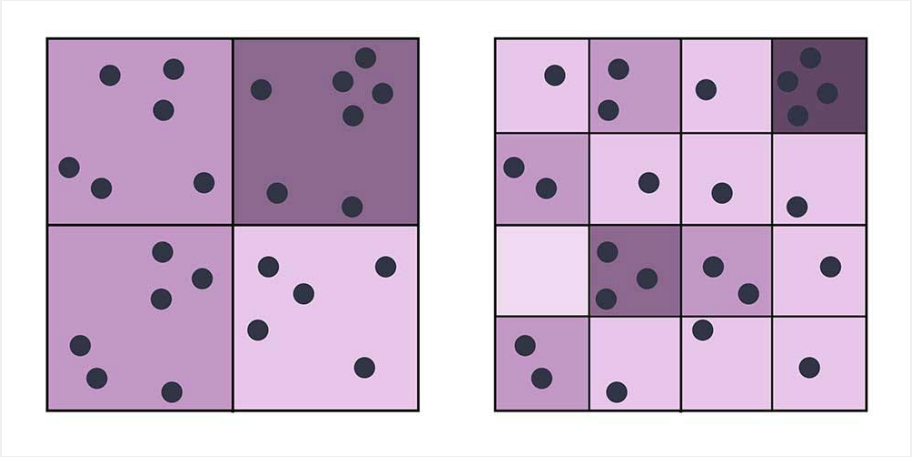
$$H = \sum_{s} p_s \ln\left(\frac{1}{p_s}\right)$$

avec p_s = proportion/probabilité d'une catégorie

Quantifier la « diversité urbaine » (II)

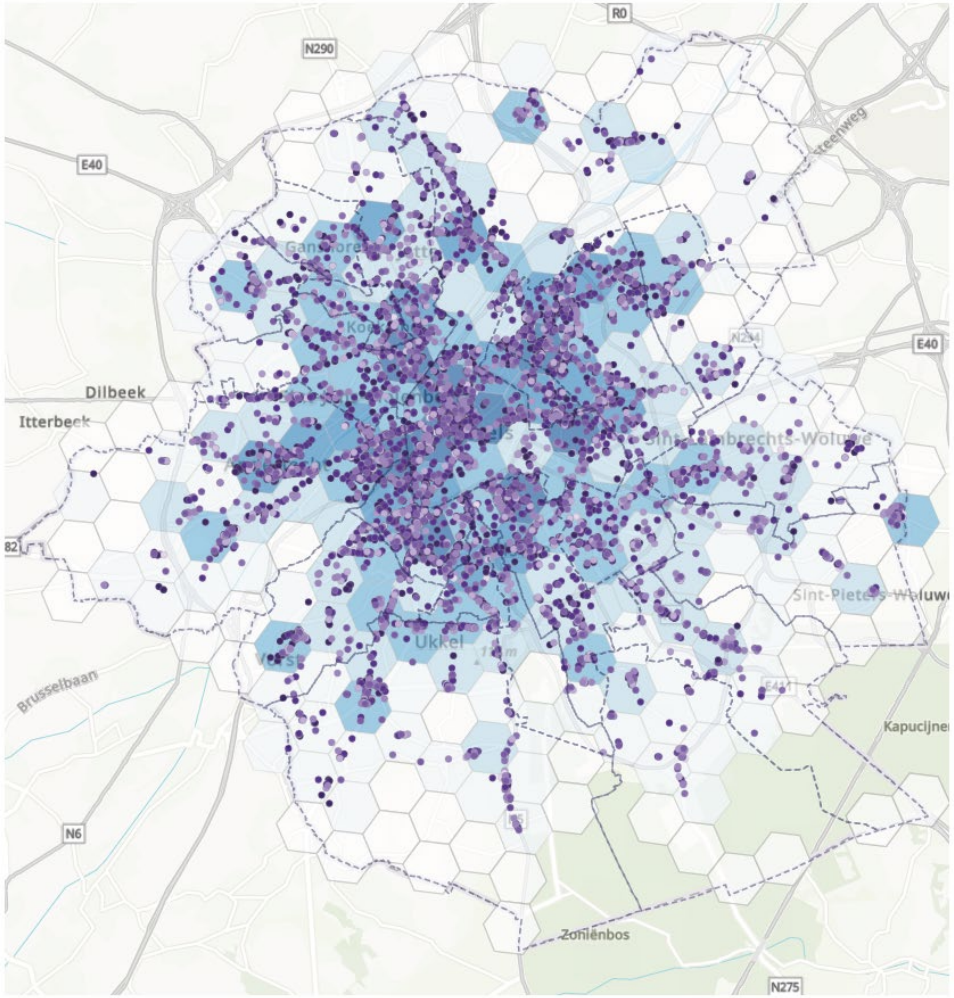
- Problèmes :

1. Mesures dépendent du découpage géographique (**MAUP**)



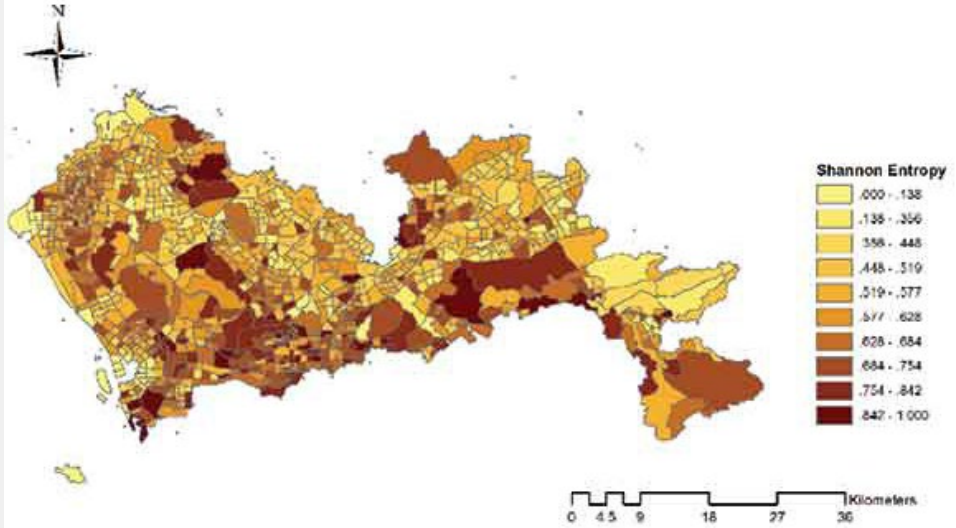
2. Mesures d'entropie sont difficiles à interpréter

CARTE 12 Analyse des équipements dans le domaine des besoins quotidiens

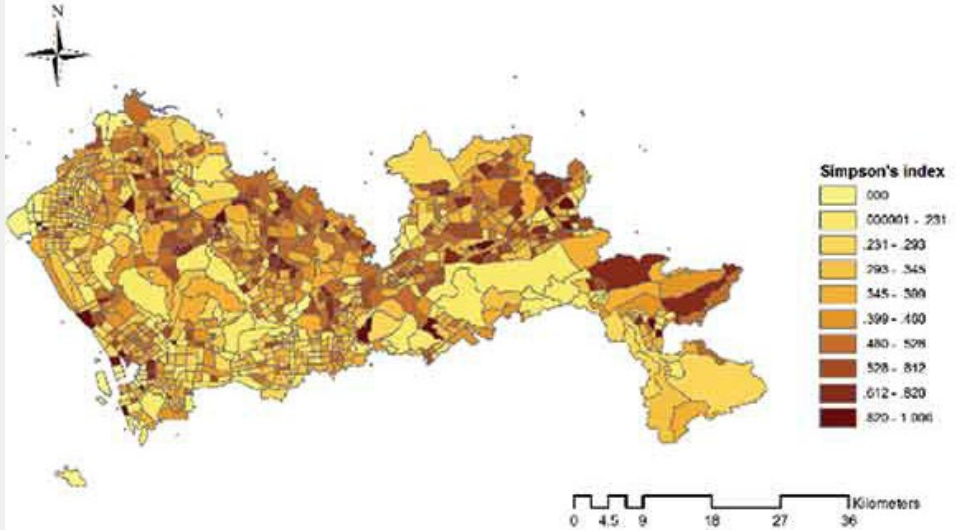


Source : Carte ArcGIS, analyse ville à 15 minutes - domaine des besoins quotidiens





b)



c)

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Measurements of POI-based mixed use and their relationships with neighbourhood vibrancy

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ABSTRACT

Although mixed use is an emerging strategy that has been widely accepted in urban planning for promoting neighbourhood vibrancy, there is no consensus on how to quantitatively measure the mix and the effects of mixed use on neighbourhood vibrancy. Shannon entropy, the most commonly used diversity measurement in assessing mixed use, has been found to be inadequate in measuring the multifaceted, multidimensional characteristics of mixed use. And lack of data also makes it difficult to find the relationship between mixed use and neighbourhood vibrancy. However, the recent availability of new sources including mobile phone data and Point of Interest (POI) data have made it possible to develop new indices of mixed use and neighbourhood vibrancy to analyse their relationships. Taking advantage of these emerging new data sources, this study used the numbers of mobile phone users in a 24-hour period as a proxy of neighbourhood vibrancy and used POIs from a navigation database to develop a series of mixed-use indicators that can better reflect the multifaceted, multidimensional characteristics of mixed-use neighbourhoods. The Hill numbers, a unified form of diversity measurement used in ecological literature that includes richness, entropy, and the Simpson index, are used to measure the degrees of mixed use. Using such fine-grained data sets and the Hill numbers allowed us to obtain better insights into the relationship between mixed use and neighbourhood vibrancy. Four models varying in POI measurements that reflect different dimensions of mixed use were presented. The results showed that either POI density or entropy can explain approximately 1% of neighbourhood vibrancy, while POI richness contributes significantly in improving neighbourhood vibrancy. The results also revealed that the entropy has limitations as a measure for representing mixed use and demonstrated the necessity of adopting a set of more appropriate measurements for mixed use. Increasing the number of POIs has limited power to improve neighbourhood vibrancy compared with encouraging the mixing of complementary POIs. These exploratory findings may be useful for adjusting mixed-use assessments and to help guide urban planning and neighbourhood design.

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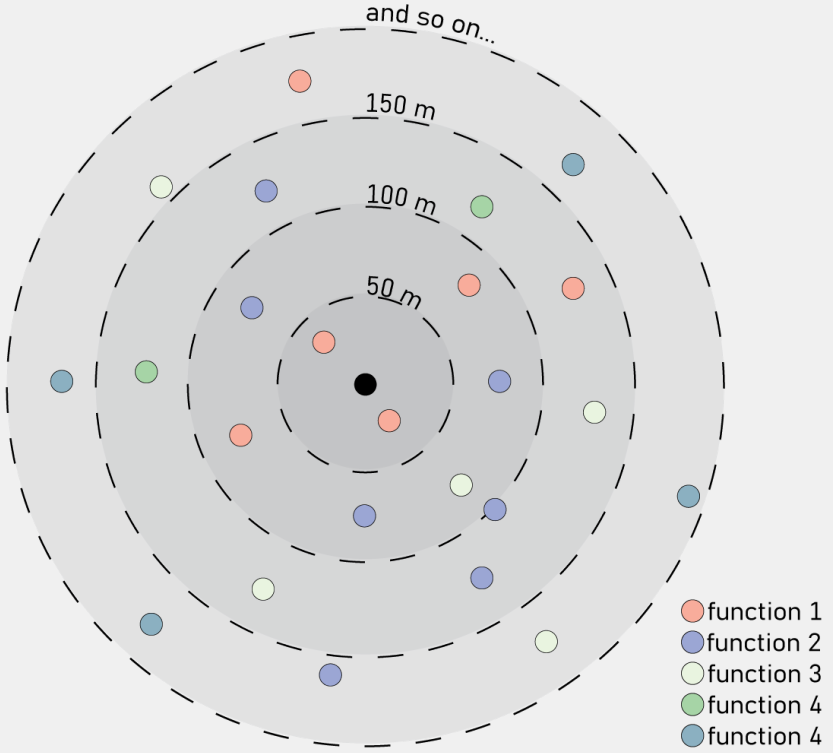
KEYWORDS
Neighbourhood vibrancy; mixed land use; diversity measurement; POI

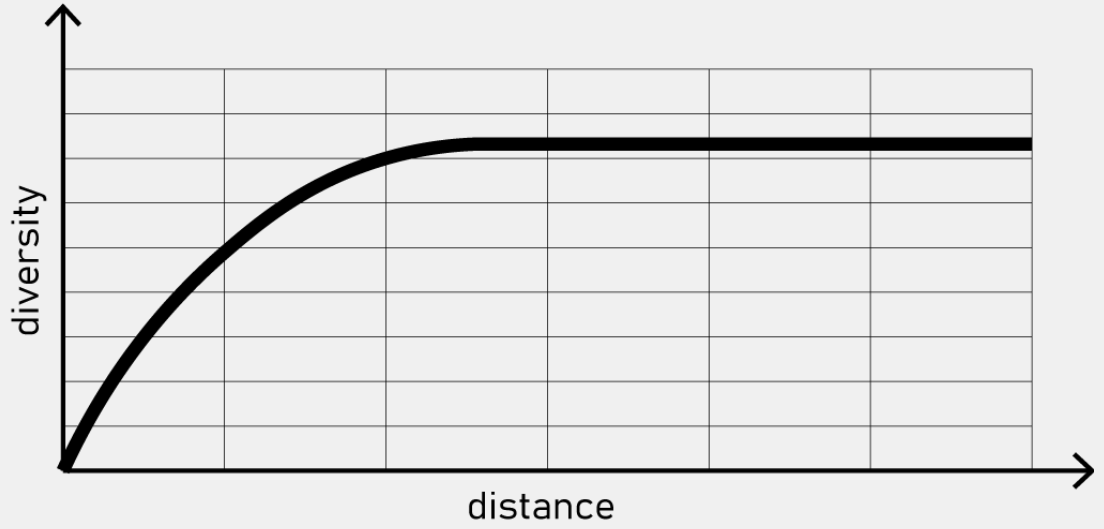
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Une autre approche (I)

- Définir des unités spatiales propres à **chaque** commerce
  - ➢ Voisinages « sur-mesure » (*bespoke neighbourhoods*)





Une autre approche (II)

- Utiliser une métrique de diversité **intelligible**
  - Nombres de Hill (1973)

$$D = exp(H) = exp\left(\sum_s p_s \, ln\left(\frac{1}{p_s}\right)\right)$$

« *the number of **equiprobable species** that obtains the same entropy as the observed distribution* » (Marcon & Puech 2025)

= le **nombre effectif** d'espèces (catégories)

Working paper
September 2025

Measuring local biodiversity with Diversity Measures at Distance

Eric Marcon[1] and Florence Puech[2]

Abstract

This study introduces a new class of biodiversity measurement - the Diversity Measures at Distance - which enables diversity evaluation at any scale of observation and provides interpretative values of the results. The new class of measures are based on geolocalized data and quantify local diversity very precisely. Diversity values are interpreted using the Hill numbers (Hill, 1973). We propose visualizing the results by plotting them in a detailed and intelligible map of the local diversity at a chosen scale. After introducing our methodology, we conduct the first application as a proof of concept, measuring the local biodiversity in the *Parc des Buttes Chaumont* in Paris. Our proposed application illustrates the great potential of this new approach and paves the way for future research. Results and maps of local biodiversity levels are provided in addition to data and R code for reproducible research.

Keywords

Diversity Measures at distance - DMD, biodiversity, spatial structure, Parisian trees, Hill numbers

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[2*]Université Paris-Saclay, INRAE, AgroParisTech, Paris-Saclay Applied Economics, F-91120, Palaiseau, France.

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Local biodiversity maps • Key results on local biodiversity provided by DMD

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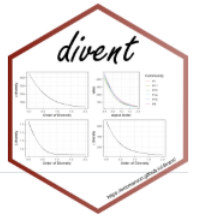
1. Introduction

Biodiversity and its conservation are a key challenge of contemporary times (IPBES, 2019; Roy et al., 2023), and evaluating biodiversity is the first step for any analysis or policy development. Therefore, measuring biodiversity precisely and making its results intelligible is essential for biodiversity analysis and the subsequent estimation of the costs associated with biodiversity erosion (Turbelin et al., 2023). Biodiversity covers all forms of living organisms; thus, many different approaches and definitions have been proposed for its evaluation. A certain number of biological diversity measures are very well-known and largely used (e.g., the Shannon index or the Simpson index) while others remain quite confidential (Krebs, 1999; Marcon, 2024).[1] While previous research has emphasized that diversity measures should be rigorously developed because each indicator responds to a particular definition of diversity, the scale at which diversity has to be evaluated is less discussed. Diversity level has generally been estimated within a predefined zone; for example, plots in forests (Broadbent et al., 2008) or regions such as the French Small Agricultural Regions (Pythagore et al., 2017; Cocco et al., 2023). To the best of our knowledge, no approach has been developed to simultaneously assess local diversity (*i*) for any distance of an environment analyzed and (*ii*) for any chosen definition of diversity. This study fills the gap by providing a new class of indicators called Diversity Measures at Distance (DMD), which are based on the Hill numbers

[1]In our study, the term "biodiversity" is used when diversity concerns a topic directly related to biology or ecology, and the term "diversity" is retained otherwise.

divent 0.5-4 Get started Reference Articles Changelog

Diversity measures based on Entropy



divent is an R package that provides functions to estimate alpha, beta and gamma diversity of communities, including phylogenetic and functional diversity.

It is a reboot of the package **entropart** to make it tidy, easier to use and optimize the code that has been added along years of research.

Installation

You can install the development version of divent from GitHub with:

```
# install.packages("pak")
pak::pak("EricMarcon/divent")
```

https://ericmarcon.github.io/divent/

Application au commerce bruxellois

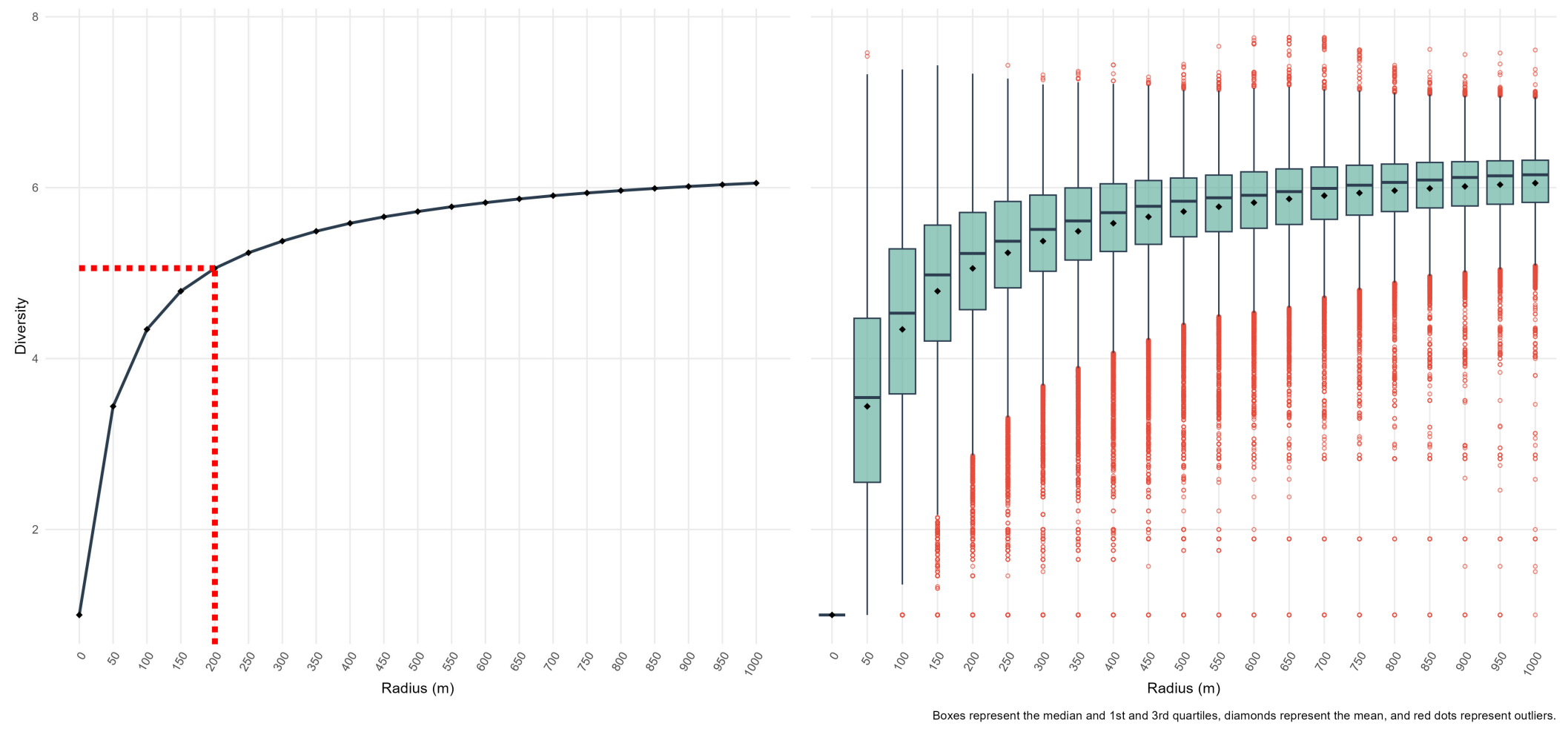
- Base de données OliGo (hub.brussels 2023)
- 3 niveaux de classification :
  - 8 catégories
  - 42 sous-catégories
  - 207 sous-sous-catégories

Application au commerce bruxellois

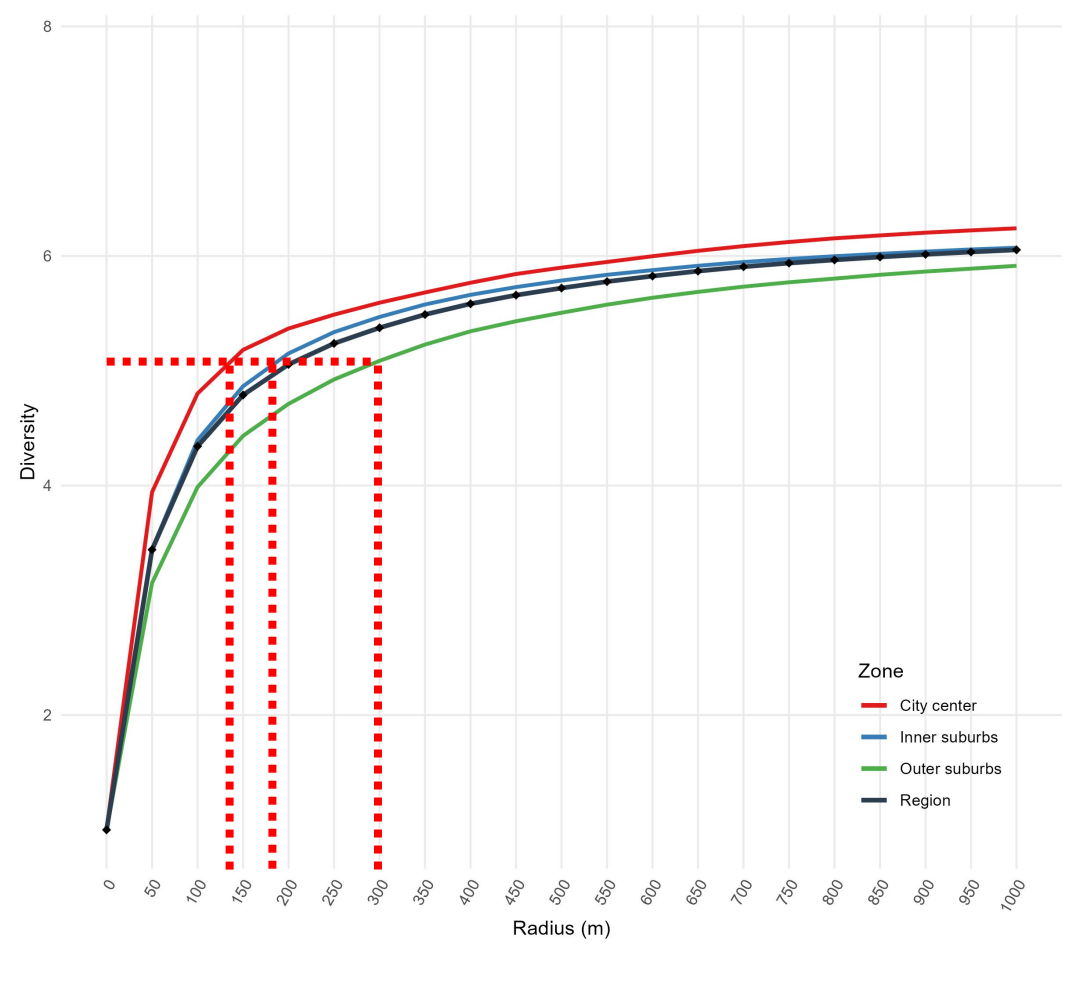
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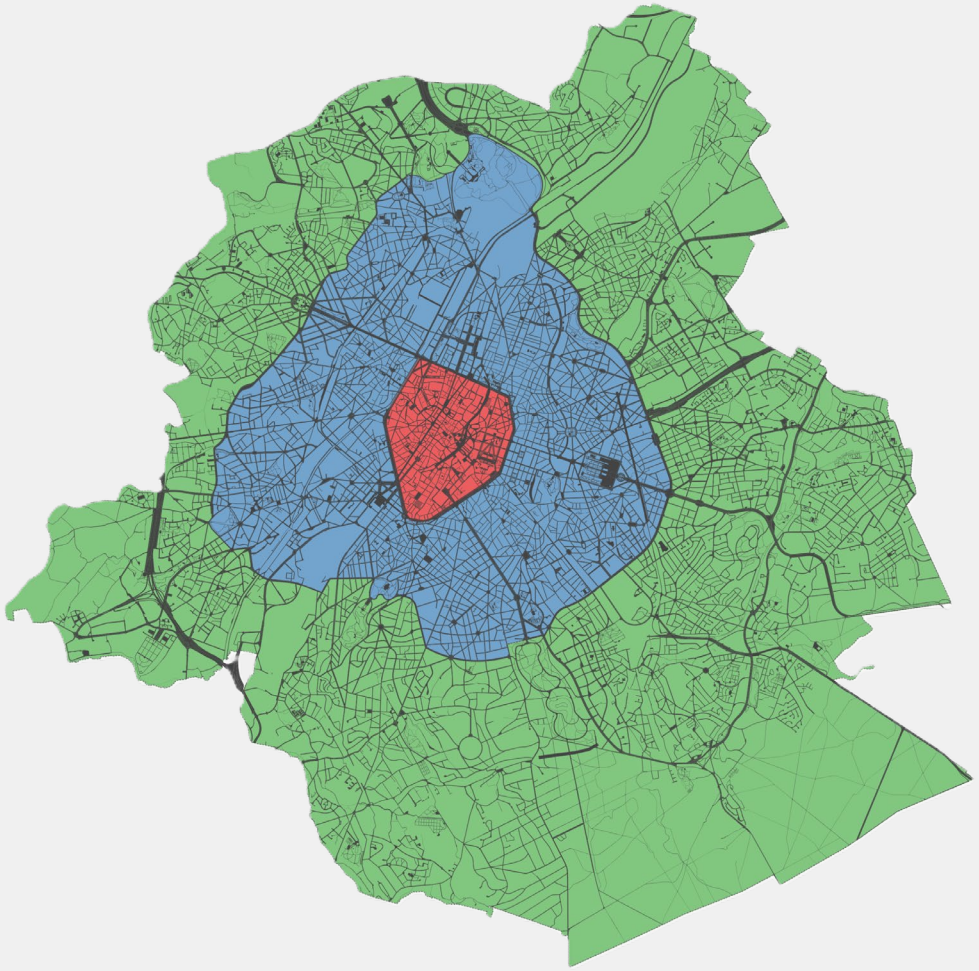
- La granularité de la classification influence les résultats !

Diversité commerciale (42 catégories)

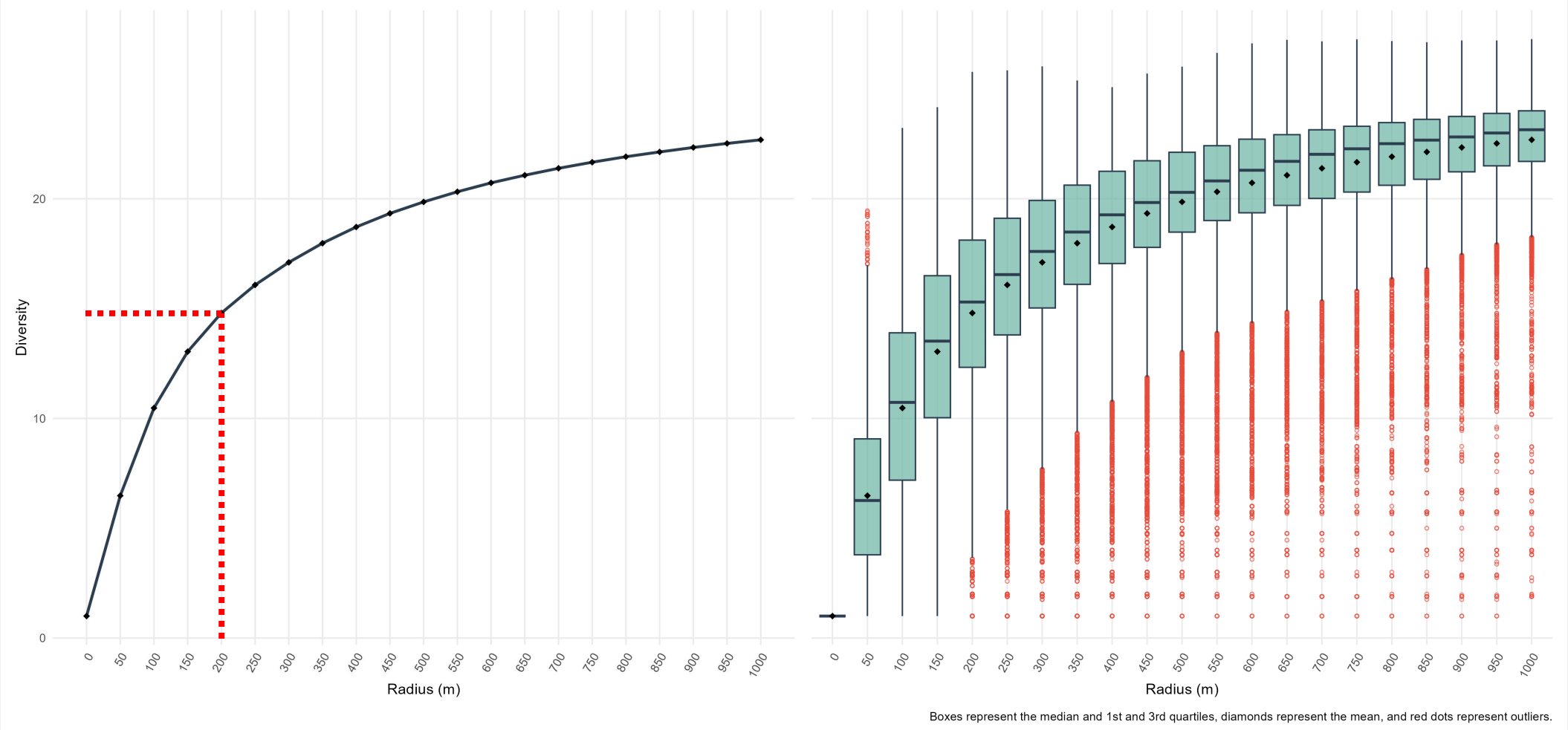


Diversité commerciale (42 catégories)

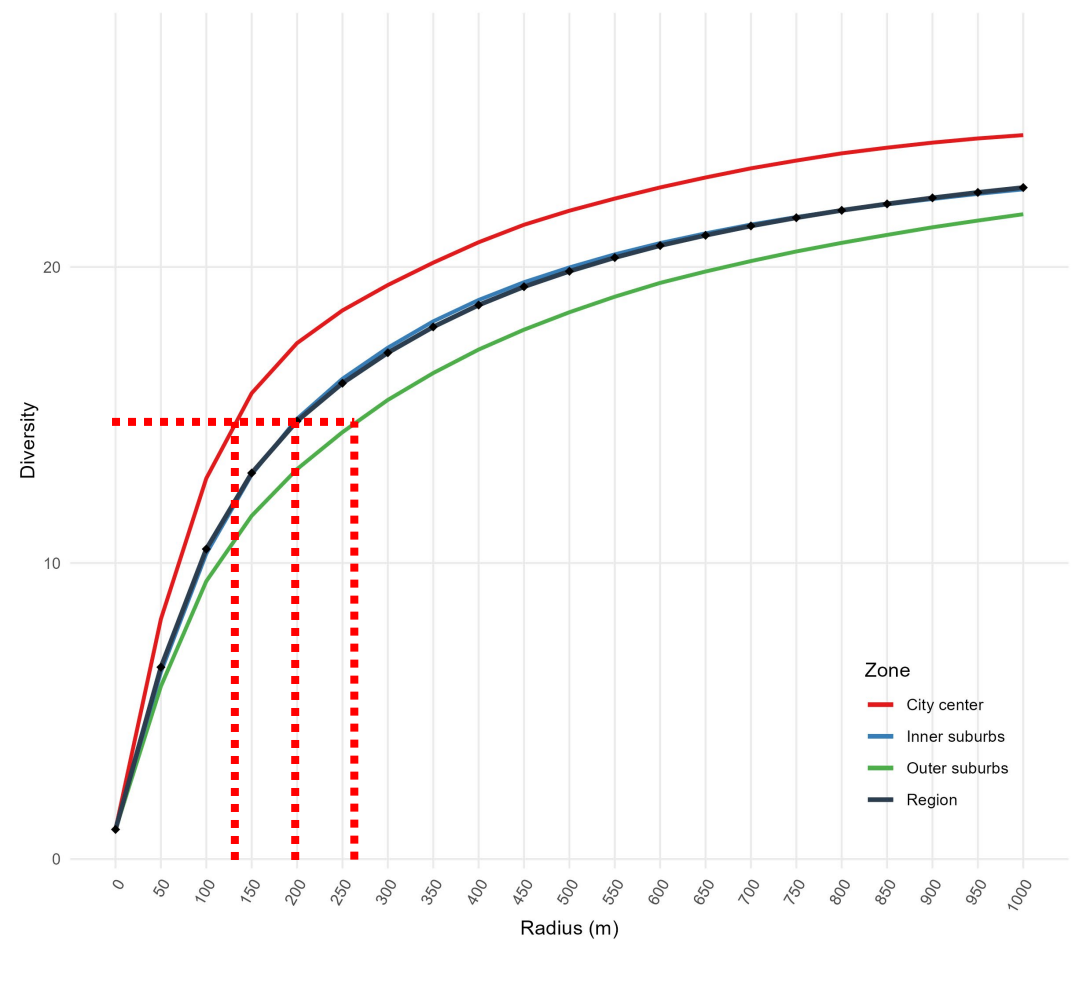


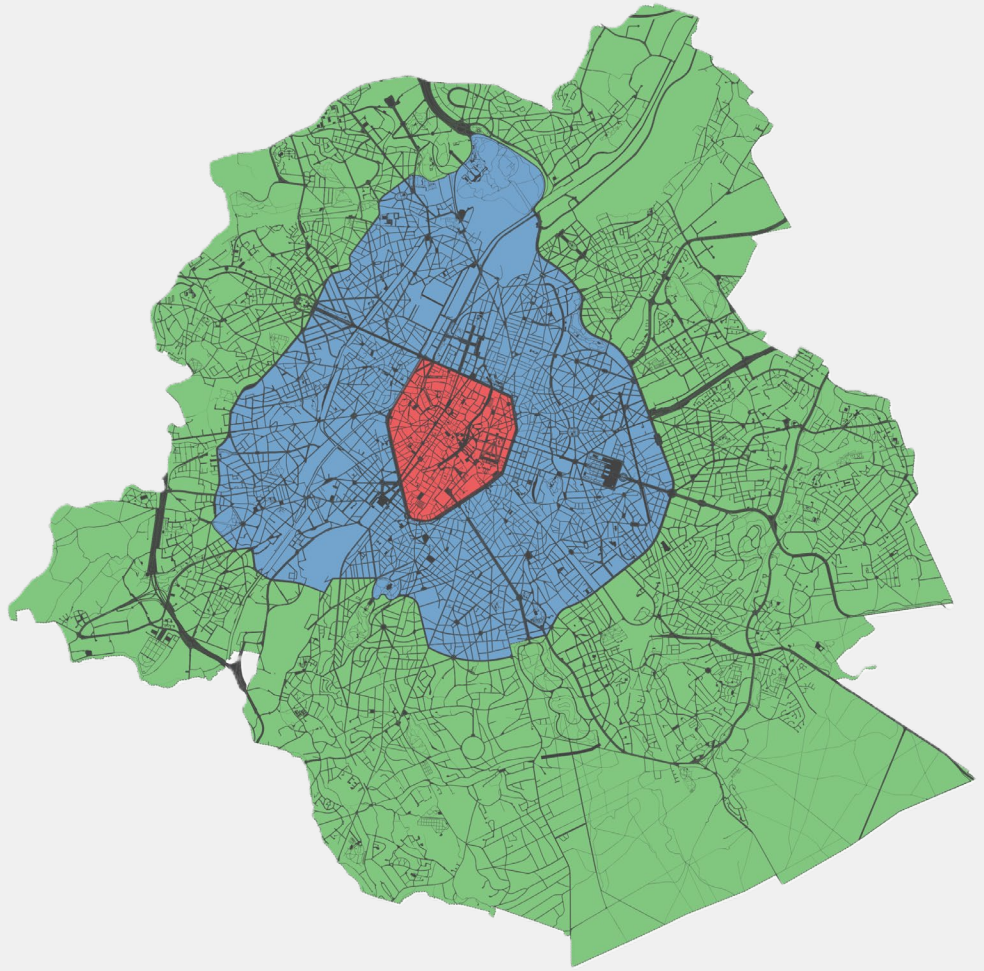


Diversité commerciale (207 catégories)



Diversité commerciale (207 catégories)

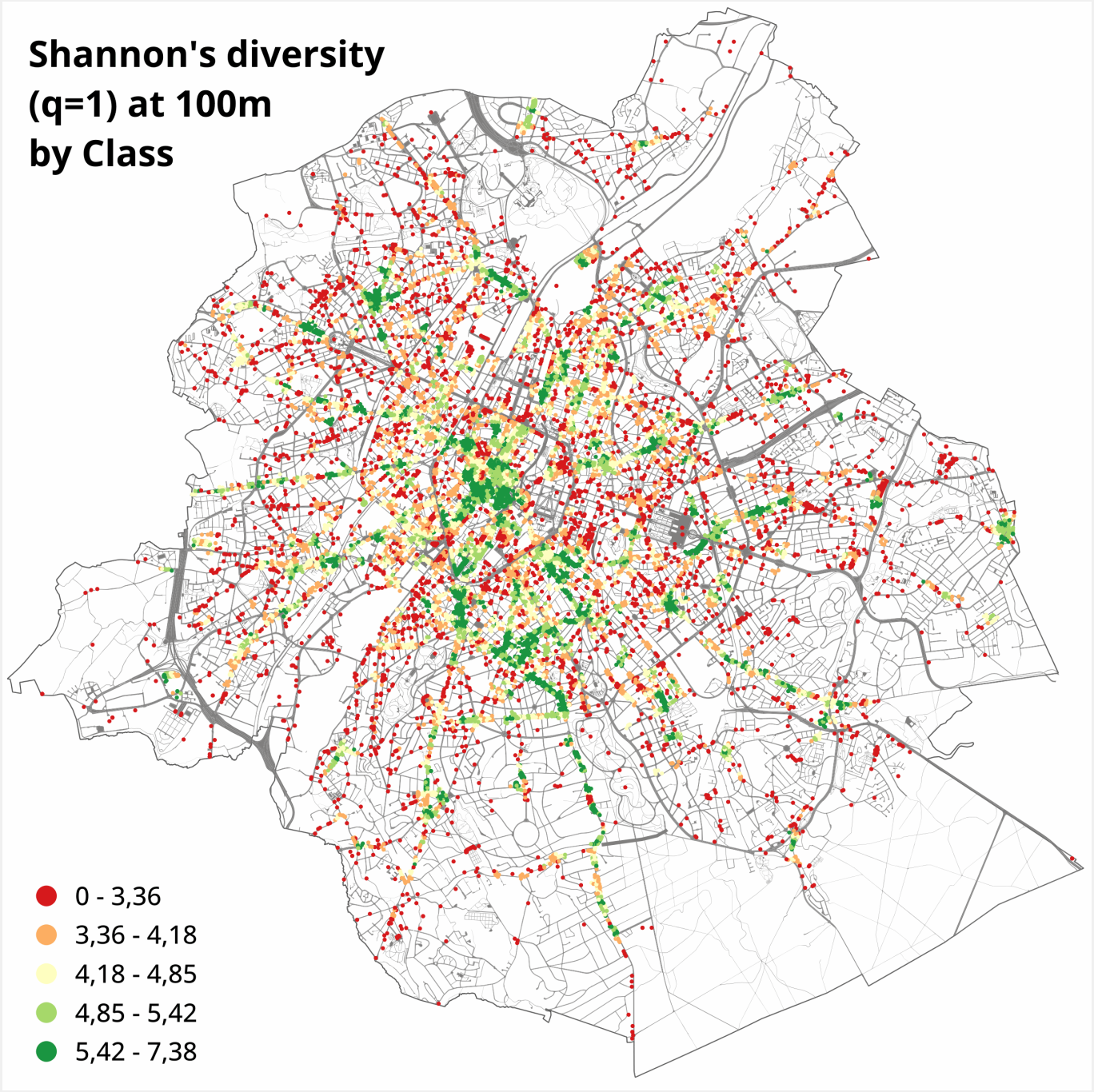


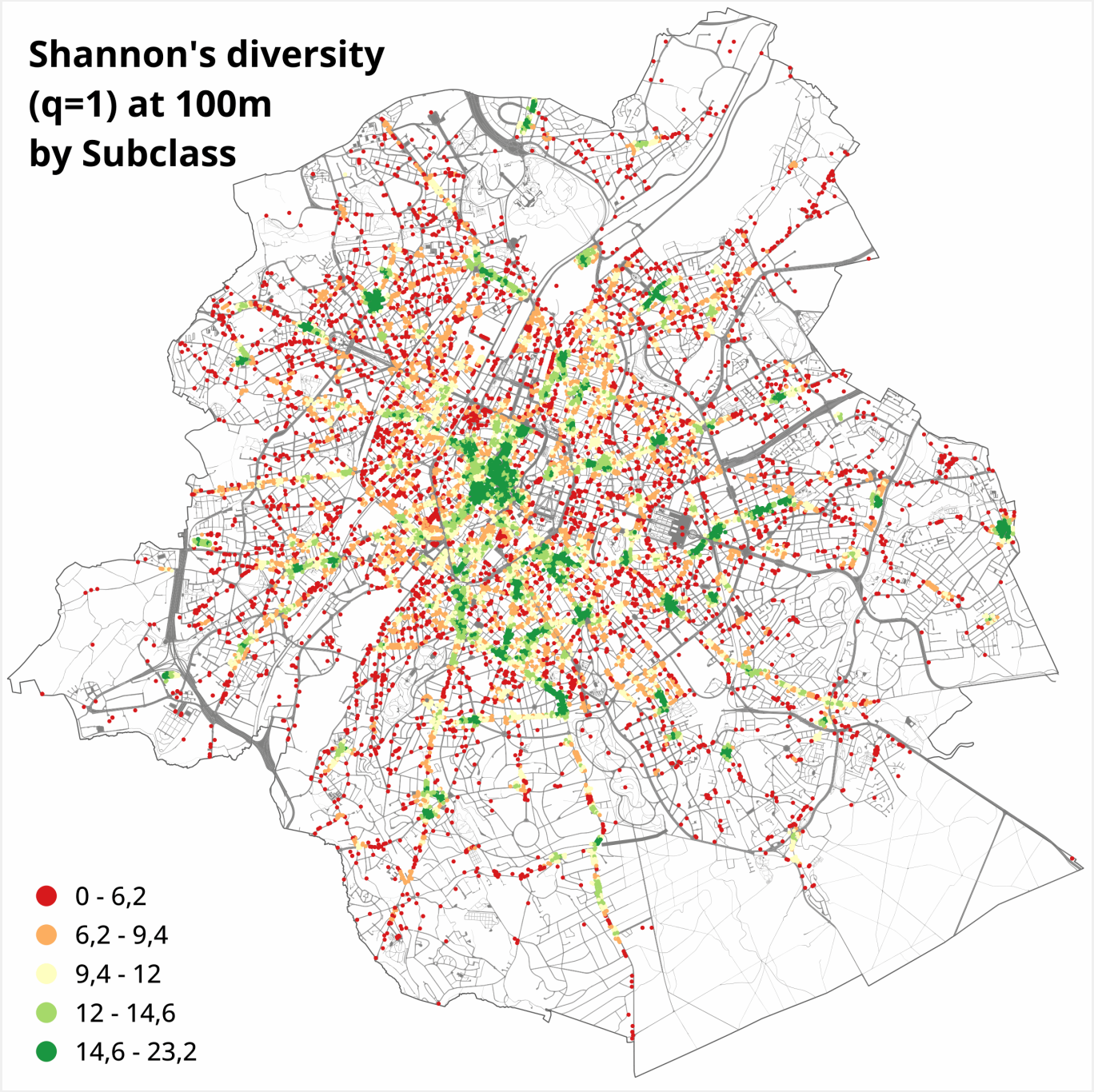


Comment cartographier ? (I)

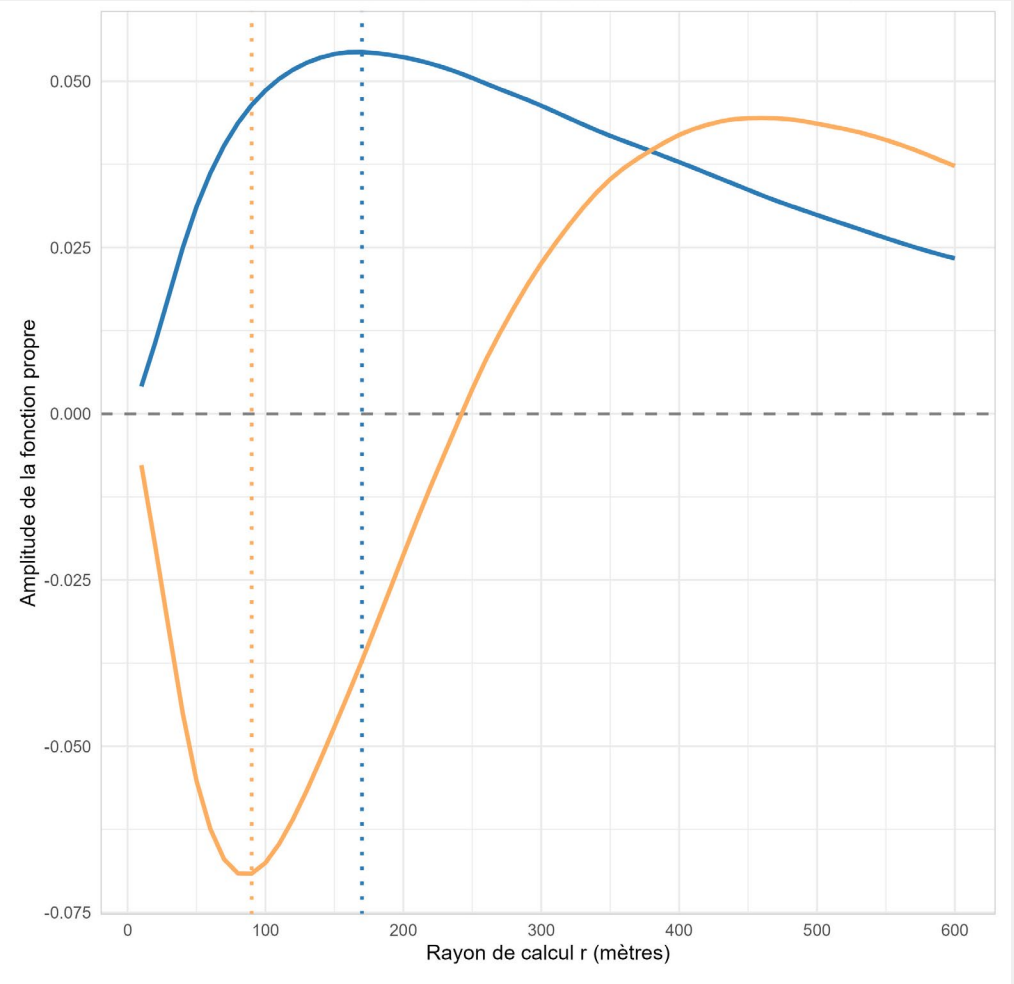
1. Fixer le niveau de diversité, représenter les rayons de calcul correspondants

2. Fixer le rayon de calcul, représenter les niveaux de diversité correspondants


Shannon's diversity
(q=1) at 100m
by Class
0 - 3,36
3,36 - 4,18
4,18 - 4,85
4,85 - 5,42
5,42 - 7,38


Shannon's diversity
(q=1) at 100m
by Subclass
0 - 6,2
6,2 - 9,4
9,4 - 12
12 - 14,6
14,6 - 23,2

Comment cartographier ? (II)



Stay tuned…

Pour la suite…

- Utiliser des distances sur réseau
- Pondérer par la surface commerciale
- Mesurer à partir d'autres points (non-commerciaux)

Scale matters!

Merci de votre attention

Hugo Colicchia

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Measuring information (I)

- Information entropy (Shannon 1948)

$$H = \sum_s p_s \ln\left(\frac{1}{p_s}\right)$$

where $\ln\left(\frac{1}{p_s}\right)$ is the amount of information

Measuring information (II)

- Generalised entropy (Tsallis 1994; Marcon et al. 2014)

$$^{q}H = \sum_{s} p_s ln_q\left(\frac{1}{p_s}\right)$$

where $\begin{cases} ln_q(x) = \frac{x^{1-q}-1}{1-q}, q \neq 1 \\ ln_q(x) = \ln(x) \end{cases}$ (deformed natural logarithm)

Measuring information (III)

- Richness (number of species/land uses minus 1):

$$^{0}H = \sum_{s} p_s \left(\frac{1}{p_s} - 1 \right)$$

- Shannon's entropy:

$$^{1}H = \sum_{s} p_s \, ln \left(\frac{1}{p_s} \right)$$

- (Gini-)Simpson's entropy:

$$^{2}H = 1 - \sum_{s} {p_s}^2$$

(cf. Marcon & Puech 2025)

Measuring diversity (I)

- Hill numbers (Hill 1973; Marcon & Puech 2025)

$$^{q}D = exp_q(^{q}H)$$

where $\begin{cases} exp_q(x) = (x(1-q)+1)^{\frac{1}{(1-q)}}, q \neq 1 \\ exp_q(x) = exp(x) \end{cases}$ (deformed exponential)

Measuring diversity (II)

- $q = 0$:

$$^{0}D = exp_0(^{0}H) = S$$

- $q = 1$:

$$^{1}D = exp_1(^{1}H) = exp\left(\sum_s p_s \, ln\left(\frac{1}{p_s}\right)\right)$$

- $q = 2$:

$$^{2}D = exp_2(^{2}H) = \frac{1}{\sum_s {p_s}^2}$$

(cf. Yue et al. 2017)

Sensitivity to rare land uses

- $q = 0$ assigns **high importance** to rare land uses
- $q = 1$ assigns the **same importance** to all land uses
- $q = 2$ assigns **almost no importance** to rare land uses